CSC-125-201: Python Programming

Credits: 3 | **Semester:** Spring 2026

Course Schedule: Tuesdays, 6:00 PM - 8:45 PM | Charlestown Campus, Building D, Room D102A

HYBRID FORMAT: Alternating IN-PERSON and VIRTUAL weeks

FIRST CLASS on February 3rd, 2026 will be IN-PERSON

Instructor: Johnathan Oneal | **Email:** johnathan.oneal@bhcc.edu

Course Description

This is an introductory course designed for any student interested in learning computer programming concepts and hands-on computational thinking, all in the context of the Python programming language. No prior programming experience is necessary. Students will use their problem-solving abilities to implement programs in Python. This course will show students how to create basic programming structures including decisions and loops. Furthermore, students will explore unique Python data structures such as tuples and dictionaries. Students will also learn to perform basic debugging techniques. At the end of this course, students will have learned enough computer science and programming concepts to write Python programs to solve problems on their own. This course will prepare students to move on to the Advanced Python Programming course.

Learning Objectives

By the end of this course, you will be able to:

- Create and execute Python programs using appropriate syntax and structure
- Declare and manipulate variables of different data types (strings, integers, floats, booleans)
- Implement decision-making logic using conditional statements (if, elif, else)
- Design and implement repetitive processes using for and while loops
- Create and call functions with parameters and return values
- Work with Python data structures including lists, tuples, and dictionaries
- Import and utilize Python standard library modules
- Implement basic error handling using try/except blocks
- Apply systematic debugging techniques to identify and resolve programming errors
- Design and complete an independent programming project from concept to implementation

Required Materials

Platform: Kaggle Notebooks (free cloud-based platform: no software installation required)

Textbooks and Resources: This course is designed to be completely free and accessible to all students. With the wealth of high-quality programming resources available online, we will use carefully selected free tutorials, documentation, and learning materials throughout the semester. All required readings and supplemental materials will be posted to the course site one week in advance, ensuring you have adequate time to access and review the content. No textbook purchases are required, the goal is to eliminate any possible barriers while providing excellent educational resources.

Students are **strongly** encouraged to seek out additional learning materials beyond AI-generated content. While AI tools can be helpful for quick answers, they are not designed as structured teaching resources. Developing the ability to identify and engage with high-quality instructional materials, particularly those aligned with your specific interests and learning goals, is a crucial skill for lifelong learning in programming and technology.

Primary Resource Sources:

- LearnPython.org: Interactive programming tutorials and exercises
- Python 101 (python101.pythonlibrary.org): High-quality Python reference and examples
- Full Stack Python (fullstackpython.com): Professional applications and career context
- Official Python documentation and community resources

Hardware Requirements:

- Computer or tablet with reliable internet access
- Web browser (Chrome, Firefox, Safari, or Edge)

Teaching Methods

This course uses a **hybrid approach** with **Tuesday classes alternating between IN-PERSON and VIRTUAL every other week, starting with IN-PERSON on February 3rd, 2026.** Each class session provides instruction on new programming concepts, demonstrations, and periodic assessments.

Weekly Schedule:

- **Tuesday (alternating):** IN-PERSON or VIRTUAL class for new concept instruction and periodic assessments
- **Monday (following week):** Homework assignment deadline at 11:59 PM

You'll work with .ipynb notebook files containing exercises and complete assignments in the cloud environment. The course emphasizes learning by doing, with extensive practice exercises and real-world applications.

Office Hours are available by appointment through email.

Student Success Philosophy

I believe in rewarding effort and genuine engagement with the material. I am committed to helping students who demonstrate they are putting in the work and making an honest attempt to learn. However, I cannot provide meaningful assistance to students who consistently submit late work or assignments that show little to no effort. Students are encouraged to reach out if they are struggling or have questions about the course material. I also value and encourage curiosity, students who ask thoughtful questions, explore beyond the basic requirements, or show genuine interest in learning will find that enthusiasm recognized and supported.

Grading and Assessment

Grade Distribution

Component	Weight	Description
Homework Assignments	25%	Weekly notebook exercises (two lowest scores dropped)
Participation	5%	In-class engagement and bi-weekly forum discussions
Pseudocode/Program Design	10%	Two algorithm design assignments
Quizzes	20%	Periodic assessments during class sessions (lowest score dropped)
Midterm Exam	20%	Exam covering Weeks 1-8 (Week 9, date TBA)
Final Project	20%	Independent programming project

Grading Scale

Grade	Numerical Range
A	94 - 100
A-	90 - 93
B+	87 - 89
B	83 - 86
B-	80 - 82
C+	77 - 79
C	70 - 76
D	60 - 69
F	0 - 59

Assignment Details

Homework Assignments (25%):

- Due the **following Monday at 11:59 PM** (except weeks with pseudocode projects)
- Submitted via file upload through Moodle
- **Two lowest homework scores will be dropped**
- **Late Policy:** 10% penalty per day for up to three days, no submissions accepted after three days

Participation (5%):

- Participation grade consists of two components: in-class engagement and bi-weekly forum discussions
- **In-Class Participation:** Active engagement during Tuesday sessions, including asking questions, contributing to discussions, and participating in collaborative activities
- **Bi-Weekly Forum Discussions:** Thoughtful contributions to online discussion forums posted every other week on Moodle. Forums will be posted on alternating weeks (Weeks 2, 4, 6, 8, 10, 12, 14). Each forum post should demonstrate engagement with course material and thoughtful reflection on programming concepts
- Forum posts due by the following Monday at 11:59 PM
- Quality over quantity: thoughtful, substantive contributions are valued over brief or superficial responses

Pseudocode/Program Design Projects (10%):

- Two algorithm design assignments replacing traditional homework
- Week 6: Design algorithm for list manipulation problem using pseudocode
- Week 12: Design algorithm for the final project (data processing using dictionaries) in pseudocode
- Focus on problem-solving and logical thinking before coding
- Due the following Monday at 11:59 PM, will replace the homework assignment for the week
- **Late Policy:** Same as homework (10% per day for up to three days, then zero)

Quizzes (20%):

- Administered during Tuesday class sessions (both in-person and virtual) periodically throughout the semester
- Cover material from previous weeks
- Focus on conceptual understanding and code reading
- Designed to ensure students are engaging with course materials appropriately and completing the homework. These are designed not to be challenging if you complete all the materials and homework for the previous week
- **Lowest quiz score will be dropped**
- No make-up quizzes except for excused absences

Midterm Exam (20%):

- Covers material from Weeks 1-8
- Scheduled for Week 9 (date to be announced)
- Mix of multiple choice, short answer, and code writing (by hand)

Final Project (20%):

- Overarching programming project combining all course concepts
- Includes planning documentation and working code
- Technical design due Week 12
- Final project due Week 15

Detailed Course Schedule

Week	Topic	Programming Concepts	Assessment
1	Setup, Intro, Syllabus Review, Python Intro	Course overview, syllabus, Python introduction, basic concepts	Homework
2	Variables and Data Types	Variables, int/float/str/bool, type conversion, input(), arithmetic	Homework + Forum
3	String Operations & Debugging	String methods, slicing, f-strings, error messages, debugging	Homework
4	Math Operations and Operators	Arithmetic/comparison/assignment operators, boolean expressions, precedence	Homework + Forum
5	Conditional Logic	if/elif/else, logical operators, nested conditions, boolean algebra	Homework
6	Lists and Tuples	Creating sequences, indexing, slicing, mutable vs immutable, basic methods	PSEUDOCODE + Forum
7	Advanced List Operations	List methods, copying, split()/join(), list comprehensions	Homework
8	For Loops	For loop syntax, range(), nested loops, enumerate(), iteration patterns	Homework + Forum
9	While Loops & Advanced Debugging	While loops, break/continue, input validation, debugging strategies	Homework
10	Midterm + Project Planning	Exam (Weeks 1-8), software development lifecycle, project scoping	**MIDTERM** + Forum
11	Dictionaries	Dictionaries, key-value pairs	Homework
12	Functions	def keyword, parameters/arguments, return values, scope, docstrings	**PSEUDOCODE** + Forum
13	Libraries and Modules	Import statements, random/math/datetime modules, documentation	Homework
14	Error Handling	try/except blocks, exception types, debugging methodology	Homework + Forum
15	Final Project	Project integration, documentation, testing, code quality	**FINAL PROJECT**

Course Policies

Academic Integrity

Students are expected to uphold the highest standards of academic honesty throughout this course. All work submitted must represent your own understanding and effort. While collaboration and discussion with classmates are encouraged for learning purposes, all submitted assignments must be your own individual work. Copying code from classmates, online sources, or any other external source without proper attribution constitutes academic dishonesty.

Students who violate academic integrity policies may receive a failing grade for the assignment or course, depending on the severity of the violation.

AI Usage Policy

Regardless of what tools or resources you use to assist in your learning, you are responsible for writing and understanding all code you submit. You must be able to explain any code you turn in and demonstrate your comprehension of how it works. If you cannot explain how your submitted code functions, it will be considered a violation of academic integrity.

Attendance and Participation

Regular attendance at Tuesday class sessions (both in-person and virtual) is required for live instruction, periodic assessments, and in-class participation credit. Active engagement during class sessions and thoughtful contributions to bi-weekly forum discussions are essential components of your learning experience and will contribute to your participation grade. The Bunker Hill Community College Attendance Policy is as follows: Because poor attendance generally results in poor grades, students must attend all regularly scheduled classes and laboratory sessions. Students having attendance difficulties should discuss this matter with their course instructors and advisors.

Students who know they will be absent for three or more consecutive class sessions because of a family emergency or personal illness should report the extended absence to the Office of the Dean of Students. The Associate Vice President will notify the course instructors of the absence. Upon returning to class, students are responsible for discussing completion of all course requirements with their course instructors. If the course instructor and student find it impossible to complete all assigned work, the student may need to withdraw from the course.

Communication

Primary communication via email and the course Moodle. Check email and course announcements regularly.

Accommodations

Academic accommodations are handled through the Disability Support Services office. Students who have questions about how to register with the Disability Support Services Office can walk in to E222 to meet with an available DSS staff member or can call or email to schedule an appointment. To schedule an appointment, contact Disability Support Services at disabilitysupport@bhcc.edu or 617-228-2327.

Distance Education Information

Course Delivery

Hybrid format: Tuesday class sessions alternate between IN-PERSON and VIRTUAL every other week

Virtual Class Meeting Information

Meeting Link: https://bhcc.webex.com/bhcc/j.php?MTID=md9efa0784e9e8c08cf3f14fd68ac6218

Schedule: Every other Tuesday, 6:00 PM - 8:45 PM (starting February 10, 2026)

Meeting Number: 2300 646 6465

Note: The meeting password will be provided via Moodle and your email invitation.

Materials Distribution

All course materials will be distributed through BHCC Moodle. This includes:

- .ipynb notebook files for exercises and assignments
- Supplemental reading materials and resources
- Assignment instructions and rubrics
- Course announcements and updates
- Bi-weekly discussion forum prompts

Technology Requirements

- Stable internet connection for virtual class sessions and Moodle access
- Web browser capable of running Kaggle Notebooks and accessing Moodle
- Email access for course communications
- Webcam and microphone for virtual class sessions

This syllabus is subject to change. Students will be notified of any modifications via course announcements.